1 Introduction

We begin by revisiting a fundamental decision-making problem, and how it is treated within the framework of Bayesian decision theory.

Consider an environment where an agent must select an action from a set of available actions. The loss associated with each action taken by the agent depends on the unknown state of the environment and can be quantified by a real number. Specifically, let a denote an action, and x represent the unknown environment state. The loss function $\ell(a, x)$ quantifies the loss for each action.

Moreover, we assume that an evidence e of this unknown environment state exists, and the agent can leverage this evidence to determine the best action. In other words, our agent aims to find the optimal strategy, where a strategy defines the distribution of each action conditioned on the observed evidence: $S(A \mid E)$.

According to Bayesian decision theory, the optimal strategy is the one that minimizes the expected loss:

$$S_{\text{opt}} = \arg\min_S \sum_a \sum_{x,e} \ell(a, x) S(a \mid e) P^*(x, e).$$

It can be shown that there always exists a deterministic optimal strategy. That is, we can consider a strategy as a function of the evidence, rather than as a conditional distribution:

$$S_{\text{opt}} = \arg\min_S \sum_{x,e} \ell(S(e), x) P^*(x, e).$$

The optimal strategy, for each evidence e, chooses the action that minimizes the expected loss conditioned on e:

$$S_{\text{opt}}(e) = \arg\min_a \mathbb{E}_{P^*} \left[\ell(a, x) \mid e\right] \tag{1}$$

$$= \arg\min_a \sum_x \ell(a, x) P^*(x \mid e)\cdot \tag{2}$$

This implies that the only distribution which affects the optimal strategy is $P^*(X|E)$. Note that the value of the expected loss depends on $P^*(E)$, but the actions that construct the optimal strategy are not affected by $P^*(E)$. Only for suboptimal strategies $P^*(E)$ plays a role and determines the importance of suboptimal actions.

Here, $P^*(X, E)$ is the joint probability distribution of the environment state and the evidence. However, how do we obtain this distribution? Is there an objective distribution inherent in the environment that we need to somehow discover?

Answering this question is not straightforward. From a subjective perspective on probability, the answer is negative. There is no such objective distribution, and P^* exists only in our minds. It serves as a tool that helps us regulate our reasoning—an intermediate measure created within the Bayesian decision-making framework to ensure we stay consistent with our assumptions and common sense principles.

Considering the subjective viewpoint, the determination of P^* relies on one's prior knowledge and assumptions. When we have no assumptions and no prior knowledge, the set of candidate probabilities for P^* simply consists of all possible probability measures. As we make assumptions or obtain some prior knowledge, those probability measures that do not align with our assumptions or knowledge are eliminated from the set. Consequently, the set of acceptable probability measures becomes more refined. We will refer to this refined set as a probabilistic model:

Definition 1.1 (Probabilistic Model)**.** We define a *probabilistic model* $\mathcal{M}$ as a pair $(\mathcal{S}, \mathcal{P})$, where $\mathcal{S}$ is a sample space, and $\mathcal{P}$ is a set of acceptable probability measures on $\mathcal{S}$.

In the statistical modeling, one of the most commonly used assumptions, that determines the set of acceptable probability measures, is the exchangeability assumption. We say a sequence of random variables $X_1, X_2, \ldots$ is exchangeable, if the ordering of variables does not provide any useful information. In other words, for an exchangeable sequence of random variables, the joint probability is the same for any permutation of variables. This property holds true for every subset of the sequence.

For example, if each X_i represents a binary variable, the joint probability depends solely on the total count of 1s and 0s, and the specific order in which these variables appear becomes irrelevant.

When a sequence of random variables is exchangeable, de Finetti's theorem states that their joint probability has a special representation:

$$P(x_1, x_2, \ldots) = \int \prod_i \pi(x_i \mid \theta) p(\theta) d\theta.$$

This means, exchangeable observations are conditionally independent relative to some latent parameter θ.

Moreover, the theorem states that if each X_i is discrete, the distribution $\pi(x_i \mid \theta)$ corresponds to the multinomial distribution.

Definition 1.2 (Parametric Model)**.** Let Θ be a real vector. A *parametric model* is a probabilistic model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$, where for every $P \in \mathcal{P}$ there is a prior probability distribution p, such that for every $s \in \mathcal{S}$

$$P(s) = \int \pi(s \mid \theta) p(\theta) d\theta, \tag{3}$$

where π is a fixed, known probability distribution. We refer to Θ as the *model parameter* and π as the *likelihood* distribution.

Now, consider our decision-making problem: Suppose the evidence E consists of a sequence of previous experiences of the environment. If we assume that our environment remains unchanged over time so that this sequence is exchangeable, to fully determine P^*, we only need to determine the prior distribution of the latent parameter $p(\theta)$.

Clearly, one approach to selecting a suitable prior distribution $p(\theta)$ is to rely on our prior knowledge about the environment. But, what if such knowledge is unavailable? This is a crucial question. In Bayesian statistics, this is where non-informative priors come into play. These priors are typically chosen to be uniform or to represent a state of ignorance about the parameters.

There is also another consideration that could play a role in choosing an appropriate prior distribution. Suppose that in our initial decision making problem, we choose to use a different evidence E' instead of E. Could this change potentially improve our strategy? Within the framework of Bayesian decision theory, we can straightforwardly answer this question. To compare any two strategies, we simply compare their expected loss. The strategy with the lower expected loss is the better one.

Note that in reality we cannot freely choose any evidence we want. We can only choose evidence that is a function of the evidence provided by the environment, allowing us to determine its value after observing the original evidence. Utilizing a strategy that uses a simpler evidence, resembles feature selection in machine learning. For instance, in a learning task where our evidence is an image, we might substitute the image with the first three coefficients of its Fourier transform. Naturally, these coefficients are a function of the original evidence.

Interestingly, if we assume $E' = f(E)$, it turns out that, according to our Bayesian framework, the optimal strategy for E is always better than any strategy that uses E'. Thus, at first glance, it appears that Bayesian decision theory does not favor feature selection.

From a certain perspective, it could be argued that adopting a strategy which uses simpler evidence may lead to ignoring some parts of useful available information, and therefore is not justified by Bayesian decision theory.

However, in practice we have to use feature selection because without it the original learning task requires impractical large datasets.

The key point here is that the space of strategies using E' is a subset of strategies that use E. When we are searching for the best strategy based on E, we are implicitly evaluating all simpler forms as well. For this reason, we believe if we choose an appropriate prior distribution, when we are finding the optimal strategy, feature selection should be performed automatically within the Bayesian decision theory framework,. However, many commonly used non-informative priors, such as the flat prior, do not exhibit feature selection characteristics.

Throughout this discussion, we will explore a systematic approach for choosing prior distributions when prior knowledge about the environment is lacking. As we will see, these priors also exhibit some interesting feature selection characteristics.

2 Central Probability Measure

When we lack prior knowledge of an environment, it is still not difficult to identify some distributions that are not suitable choices for $p(\theta)$. For instance, any distribution that assigns zero probability to certain values of θ is not a good choice. If we use such a prior, evidence can never convince us that the value of θ lies within those excluded values.

We should note that, in this context, we cannot objectively assert that there is some type of underlying process in the environment for generating different values of θ that does not consider zero probability for any value of θ. Our argument is subjective. We consider some priors to be unsafe because they have some characteristics that appear to be problematic: in the obtained posterior, some values of the parameter are excluded regardless of the evidence.

We can generalize this idea and use a similar subjective approach to determine a somewhat "safe" prior distribution. Recall that our probabilistic model for the environment consists of a set of acceptable probability measures $\mathcal{P}$. Suppose that we choose a probability measure $C \in \mathcal{P}$, and determine the optimal Bayesian strategy in C. If the expected loss of this strategy in any other probability measure $P \in \mathcal{P}$ is close to the optimal strategy, we can say that using C is safe, in the sense that the optimal strategy obtained using C, is a near optimal strategy in any other acceptable probability measure in our model.

The probability measure that we choose represents our beliefs about the environment. It can be argued that this choice must depend solely on the environment. As long as the environment is the same, we should not change this choice. For example, if our loss function changes, or if the evidence that we get from the environment becomes more or less informative, none of these can change our beliefs about the environment and therefore our chosen probability measure. We will use this principle several times in this discussion and refer to it as the *principle of objectivity*.

2.1 Divergence Measure

Unfortunately, the expected loss of a strategy is a functional of the loss function. Therefore, we cannot directly use the expected loss of the optimal Bayesian strategy for evaluating different probability measures. We need a measure that quantifies the similarity between two probability measures while it ignores the specific loss function we are using.

Suppose that for an environment, we have decided on a probabilistic model $\mathcal{M}$, and without any particular prior knowledge, we have chosen a probability measure P from our model. In particular, we are interested in the distribution of a random variable X.

We ask an expert—a person familiar with the environment—to evaluate our probability measure P. Presumably, that expert has his own beliefs about the environment. Therefore, he believes, instead of P, we should use a different probability measure Q. When evaluating P, he would try to measure the loss of using the incorrect distribution $P(X)$, while he believes the correct distribution is $Q(X)$.

Let us assume that a real number can be used to measure this loss, and denote it as $\mathbb{D}\left(Q_X \parallel P_X\right)$. It seems reasonable, as P gets closer to Q this measure decreases, and increases when P diverges from Q. Therefore, $\mathbb{D}\left(Q_X \parallel P_X\right)$ effectively is a measure of the divergence of $P(X)$ from $Q(X)$, when Q is the believed true probability measure.

Interestingly, if we make certain assumptions about the properties of this divergence measure, it turns out that—up to a constant factor—it must be equal to the Kullback–Leibler (KL) divergence.

Alternatively, by adopting a more minimal set of properties, we can prove that if we accept Shannon's entropy as a measure of the uncertainty of a distribution, we should also consider KL divergence as the appropriate method for measuring divergence between distributions.

Theorem 1. *Let X be a discrete random variable. Suppose that $\mathbb{D}\left(Q_X \parallel P_X\right)$ is a divergence measure that quantifies the divergence of $P(X)$ from $Q(X)$. Consider the following properties:*

Consistency with entropy *Let Ω be a continuous random variable, and Q be a probability measure. If $U(\Omega)$ is the uniform distribution with the same support as $Q(\Omega)$, then we have*

$$\mathbb{D}\left(Q_\Omega \parallel U_\Omega\right) = \mathbb{H}\left[U_\Omega\right] - \mathbb{H}\left[Q_\Omega\right],$$

where $\mathbb{H}$ denotes Shannon's entropy.

Additivity *For any two discrete random variables X, Y, and probability measures P, Q*

$$\mathbb{D}\left(Q_{X,Y} \parallel P_{X,Y}\right) = \mathbb{D}\left(Q_X \parallel P_X\right) + \sum_x \mathbb{D}\left(Q_{Y|x} \parallel P_{Y|x}\right) Q(x)\cdot$$

If $\mathbb{D}$ satisfies these properties, then it is the Kullback-Leibler divergence.

Proof. Let $P(X)$ and $Q(X)$ be two arbitrary distributions for a discrete random variable X. We define a new continuous random variable Ω and let

$$p(\Omega = \omega \mid X = x) = \begin{cases} \frac{1}{P(x)} & 0 \leq \omega \leq P(X = x), \\ 0 & \text{otherwise}\cdot \end{cases}$$

Notice that both $P(\Omega \mid X)$ and $P(\Omega, X)$ are uniform distributions, and $\mathbb{H}\left[P_{\Omega|x}\right] = \ln P(x)$, $\mathbb{H}\left[P_{\Omega,X}\right] = 0$.

We define $Q(\Omega \mid x)$ to be an arbitrary distribution that has the same support as $P(\Omega \mid x)$. This ensures that $Q(\Omega, X)$ and $P(\Omega, X)$ have the same support as well. Thus, we have

$$\begin{aligned} \mathbb{D}\left(Q_{\Omega,X} \parallel P_{\Omega,X}\right) &= \mathbb{H}\left[P_{\Omega,X}\right] - \mathbb{H}\left[Q_{\Omega,X}\right] \\ &= \sum_x \int q(\omega, x) \ln q(\omega, x) d\omega \\ &= \sum_x Q(x) \int q(\theta \mid x) \ln q(\omega, x) d\omega, \end{aligned}$$

and

$$\begin{aligned} \mathbb{D}\left(Q_{\Omega|x} \parallel P_{\Omega|x}\right) &= \mathbb{H}\left[P_{\Omega|x}\right] - \mathbb{H}\left[Q_{\Omega|x}\right] \\ &= \ln P(x) + \int q(\theta \mid x) \ln q(\theta \mid x) d\omega\cdot \end{aligned}$$

Now, we use the additivity property to obtain a formula for our divergence measure:

$$\begin{aligned}
\mathbb{D}\left(Q_X \parallel P_X\right) &= \mathbb{D}\left(Q_{\Omega,X} \parallel P_{\Omega,X}\right) - \sum_x \mathbb{D}\left(Q_{\Omega|x} \parallel P_{\Omega|x}\right) Q(x) \\
&= \sum_x Q(x) \left(\int q(\theta \mid x) \ln \frac{q(\omega, x)}{q(\theta \mid x)} d\omega - \ln P(x) \right) \\
&= \sum_x Q(x) \left(\ln Q(x) \int q(\theta \mid x) d\omega - \ln P(x) \right) \\
&= \sum_x Q(x) \ln \frac{Q(x)}{P(x)}. \qquad \square
\end{aligned}$$

In this theorem, the first property simply states that the entropy of a distribution should be related to the divergence of the uniform distribution from it. The more the divergence is, the less the entropy must be.

The second property tells us that the divergence is an additive quantity. We know that the divergence of $P(X,Y)$ from $Q(X,Y)$ is a measure of the generic loss of using $P(X,Y)$ when we believe $Q(X,Y)$ is the true distribution. To calculate this loss, we can assume that X happens before Y. In that way, this loss would include the loss of using $P(X)$, and then, if we observe $X = x$, which can happen with probability $Q(x)$, it would also include the loss of using $P(Y \mid X = x)$. This property indicates that for combining these individual terms we should use addition.

Thus, if Q is the probability measure that represents our beliefs, we can use $\mathbb{D}\left(Q_X \parallel P_X\right)$ to evaluate any other probability measure P, with respect to a random variable X. In other words, we evaluate $P(X)$ by $Q(X)$.

If we have two random variables, X and Y, there would be two distinct cases that should not be confused: We may want to evaluate $P(X,Y)$, or we may want to evaluate $P(X)$ and $P(Y)$ separately. Evaluating the joint distribution $P(X,Y)$ is straightforward and can be done by defining a new random variable $Z = (X,Y)$.

However, the correct approach for handling the second case is not obvious. Here, we may use a similar argument to what we used for the additive property. When we want to evaluate the loss of $P(X,Y)$, If X happens first we will face the loss of $P(X)$, then when Y happens, we will face the loss of $P(Y)$ only if X and Y are independent. Therefore, evaluating the marginal distributions $P(X)$ and $P(Y)$ appears to be the same as evaluating $P(X,Y)$ when X and Y are independent, and we can use addition:

$$\mathbb{D}\left(Q_X \parallel P_X\right) + \mathbb{D}\left(Q_Y \parallel P_Y\right).$$

Separate evaluation of random variable distributions is essential for the divergence measure that we will define shortly.

The KL divergence provides a method for measuring the agreement of a model about a specific distribution. This agreement can be called convergence.

Definition 2.1 (ϵ-convergence)**.** In a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$, let X and E be discrete random variables. We say $X|E$ (X conditioned on E) *ϵ-converges by* $P \in \mathcal{P}$, if for any probability measure $Q \in \mathcal{P}$ we have:

$$\mathbb{E}_Q \left[\mathbb{D} \left(Q_{X|e} \parallel P_{X|e} \right) \right] < \epsilon,$$

where $\mathbb{E}_Q [\,\cdot\,]$ denotes expectation relative to Q and $\mathbb{D} \left(Q_{X|e} \parallel P_{X|e} \right)$ is the Kullback-Leibler divergence of $P(X \mid e)$ from $Q(X \mid e)$:

$$\mathbb{D} \left(Q_{X|e} \parallel P_{X|e} \right) = \sum_x Q(x \mid e) \ln \frac{Q(x \mid e)}{P(x \mid e)}.$$

ϵ-convergence is closely related to the concept of Bayesian convergence, where the posterior converges to the same distribution regardless of the prior distribution.

As an example, consider a scenario where we are trying to predict the outcome of a four-sided dice. This dice is unfair, with each face marked by a binary number: 00, 01, 10, and 11. Before making our prediction, we can roll the dice multiple times, with each roll being independent.

If we let E be a vector representing the first 10 dice rolls and X be the outcome of the next roll, $X|E$ ϵ-converges by P_u for $\epsilon \geq 0.109$, where P_u is the probability measure obtained from assuming a uniform prior for the multinomial parameters. If E represents the first 20 outcomes, then we can choose any $\epsilon \geq 0.073$.

Here, the chosen prior for the multinomial parameters affects the value of ϵ. In other words, different probability measures in our model have different convergence properties.

If there is a probability measure that provides ϵ-convergence for a very small ϵ, we can hardly justify using any other probability measure. However, that probability measure may not always exist. Furthermore, we cannot determine what value of ϵ would be small enough. This becomes more problematic when we need to consider more than one distribution. For example, suppose that we are interested in the distributions of both $X|E_1$ and $X|E_2$. As we discussed earlier, when dealing with multiple distributions we should consider the sum of their KL divergences. Unfortunately, that would make determining the appropriate value of ϵ even harder because that naturally increases the value of ϵ.

We could avoid all these hurdles by simpling choosing the probability measure that gives the smallest possible ϵ. Subjectively, this probability measure is the "safest" choice in our model and it turns out that it performs well even when the minimum achievable value of ϵ is not very small.

Definition 2.2 (Query Set)**.** A *query set* is a set of pairs of discrete random variables, all defined on the same sample space.

Definition 2.3 (Divergence Measure)**.** Let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$, be a query set in a probabilistic model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$. We define the *divergence measure of* $P \in \mathcal{P}$ *for* $\mathcal{X}$ as follows:

$$\mathbb{D}_{\mathcal{X}}\left[P\right] = \sup_{Q \in \mathcal{P}} \mathbb{E}_Q \left[\sum_i \mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right], \tag{4}$$

$$= \sup_{Q \in \mathcal{P}} \sum_i \sum_{x_i, e_i} Q(X_i = x_i, E_i = e_i) \ln \frac{Q(X_i = x_i \mid E_i = e_i)}{P(X_i = x_i \mid E_i = e_i)}. \tag{5}$$

Note that we usually simplify our notation and instead of writing the full assignment to a random variable, we only write the assigned value. For example, instead of $Q(X_i = x_i, E_i = e_i)$ we write: $Q(x_i, e_i)$.

Proposition. $X|E$ ϵ-converges by P, if and only if

$$\mathbb{D}_{\mathcal{X}}\left[P\right] < \epsilon,$$

where $\mathcal{X} = \{(X, E)\}$.

For a parametric model (as defined in Definition 1.2), it seems reasonable to define a conditional form of our divergence measure, conditioned on the value of the parameter.

To do so, in Equation 4, we replace normal expectation with conditional expectation. There will be no longer a need to maximize over different distributions, since in a parametric model the parameter fully determines the distribution of the sample space.

Definition 2.4 (Conditional Divergence Measure)**.** Assume that $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is a parametric model, and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ is a query set. We define the *divergence measure of* $P \in \mathcal{P}$ *for* $\mathcal{X}$ *given* θ as follows:

$$\mathbb{D}_{\mathcal{X}}\left[P \mid \theta\right] = \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \middle| \theta \right] \tag{6}$$

$$= \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi\left(x_i \mid e_i, \theta\right)}{P\left(x_i \mid e_i\right)}. \tag{7}$$

Proposition. For any value of θ, the conditional divergence is non-negative.

Now that we have established a divergence measure, we are ready to provide a formal definition for the probability measure that we consider the "safest" choice in a model.

Definition 2.5 (Central Probability Measure)**.** In a model $\mathcal{M} \equiv (\mathcal{S}, \mathcal{P})$, we define the *central probability measure* for a query set $\mathcal{X}$, to be a probability measure $C_{\mathcal{X}} \in \mathcal{P}$ which minimizes the divergence measure for $\mathcal{X}$:

$$C_{\mathcal{X}} = \arg\min_{P \in \mathcal{P}} \mathbb{D}_{\mathcal{X}} [P] \cdot$$

In this definition, the query set $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ plays a crucial role. Specifically, $\mathcal{X}$ defines the set of random variables in which we are interested. Each X_i represents a random variable whose distribution is important to us, and each E_i corresponds to the evidence that is available for that variable.

However, if we choose our query set based on our specific experimental setup, we will violate the principle of objectivity. It is important that we use the same query set for any problem that can be defined within the same environment.

One natural way to do this is to let the query set include all potentially relevant conditional pairs. For example, in our dice problem, we might let each X_i be the outcome of the i-th dice roll, and define each E_i as a vector of the previous dice outcomes: $E_i = (X_1, X_2, \ldots, X_{i-1})$. We may assume that our set contains n pairs. The choice of n should be based on our assumptions about the complexity of the problem. It shows where we think the convergence becomes trivial and, if necessary, could approach infinity.

Now let us calculate the conditional divergence measure for $\mathcal{X}$ as defined in Definition 2.4:

$$\begin{aligned}
\mathbb{D}_{\mathcal{X}} [P \mid \theta] = &\sum_{x_1} \pi(x_1 \mid \theta) \ln \frac{\pi(x_1 \mid \theta)}{P(x_1)} \\
&+ \sum_{x_2, x_1} \pi(x_2, x_1 \mid \theta) \ln \frac{\pi\left(x_2 \mid x_1, \theta\right)}{P\left(x_2 \mid x_1\right)} \\
&+ \cdots \\
&+ \sum_{x_n, x_1, \ldots, x_{n-1}} \pi(x_n, x_1, \ldots, x_{n-1} \mid \theta) \ln \frac{\pi\left(x_n \mid x_1, \ldots, x_{n-1}, \theta\right)}{P\left(x_n \mid x_1, \ldots, x_{n-1}\right)} \cdot
\end{aligned}$$

After factoring out $\pi(x_1, \ldots, x_n \mid \theta)$ from every term:

$$\sum_{x_1, \ldots, x_n} \pi(x_1, \ldots, x_n \mid \theta) \ln \frac{\pi(x_1 \mid \theta) \pi(x_2 \mid x_1, \theta) \ldots \pi(x_n \mid x_1, \ldots, x_{n-1}, \theta)}{P(x_1) P(x_2 \mid x_1) \ldots P(x_n \mid x_1, \ldots, x_{n-1})},$$

we use the chain rule for conditional probability:

$$\mathbb{D}_{\mathcal{X}}\left[P \mid \theta\right] = \sum_{x_1,\ldots,x_n} \pi(x_1,\ldots,x_n \mid \theta) \ln \frac{\pi(x_1,\ldots,x_n \mid \theta)}{P(x_1,\ldots,x_n)}. \tag{8}$$

This implies that we could alternatively define our query set as a set containing a single pair $\mathcal{X} = \{(X, E)\}$, where X represents the vector of n dice rolls $X = (X_1, X_2, \ldots, X_n)$, and E is a dummy certain event with $P(E) = 1$. This query set essentially corresponds to the entire sample set for n dice rolls. We usually simplify our notation and omit dummy events: $\mathcal{X} = \{X\}$.

The fact that our query set includes the entire sample set does not necessarily mean its central probability measure has good convergence properties for every definable random variable. Back to our dice example, suppose we are interested in predicting the next dice outcome based on the number of previous odd and even outcomes. Let Y_i represent the least significant bit of the i-th dice roll, indicating whether the number is odd or even. If we want our central probability measure to exhibit good convergence properties for $P(X_i \mid Y_1, Y_2, \ldots, Y_{i-1})$, we need to explicitly include these conditional pairs in our query set, in the same way we did for $X_i|X_1, X_2, \ldots, X_{i-1}$.

After adding these pairs, the central probability measure for this new query set shows some interesting feature selection capabilities. In this probability measure, for smaller values of i, $P(X_i \mid X_1, X_2, \ldots, X_{i-1})$ tends to get somewhat combined with $P(X_i \mid Y_1, Y_2, \ldots, Y_{i-1})$. For example when $i = 2$, if the single observed dice roll is 01, the posterior probabilities of $00, 10$ and 11 are not equal, and $P(11 \mid 01) = 0.158$ is slightly higher than $P(00 \mid 01) = P(10 \mid 01) = 0.133$. It can be said that this central probability measure identifies that 01 and 11 are both odd numbers.

2.2 Parametric Models

In a parametric model, the maximum of divergence happens for a prior distribution that is concentrated at a single value of the parameter θ. That means, for a parametric model, which allows any high peak prior, instead of maximizing over the set of all acceptable distributions, we can simply maximize over the parameter space, using the conditional version of the divergence measure.

Definition 2.6 (Generic Parametric Model)**.** We say a parametric model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is *generic*, if for any arbitrary probability distribution r, there is a probability measure $R \in \mathcal{P}$, such that for every $s \in \mathcal{S}$

$$R(s) = \int \pi(s \mid \theta) r(\theta) d\theta \cdot$$

Lemma 1. *Assume that* $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ *is a parametric model. Let* $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ *be a query set in* $\mathcal{M}_\Theta$*. For every* $P \in \mathcal{P}$ *we have*

$$\mathbb{D}_{\mathcal{X}}[P] \leq \sup_{\theta \in \Theta} \mathbb{D}_{\mathcal{X}}[P \mid \theta]$$

Proof. Let R and P be two arbitrary probability measures in $\mathcal{M}_\Theta$. For any parameter value θ we have:

$$\mathbb{D}_{\mathcal{X}}[P \mid \theta] - \mathbb{D}_{\mathcal{X}}[R \mid \theta] = \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \cdot$$

The conditional divergence is non-negative and we have

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \mathbb{D}_{\mathcal{X}}[P \mid \theta] \cdot$$

This inequality holds for every θ. Therefore, the maximum of the right-hand side w.r.t. θ is also greater than or equal to the maximum of the left-hand side:

$$\sup_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \sup_{\theta \in \Theta} \mathbb{D}_{\mathcal{X}}[P \mid \theta] \cdot \tag{9}$$

Trivially, for any θ we have

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \leq \sup_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \cdot$$

$\mathcal{M}_\Theta$ is a parametric model and $R(x_i, e_i) = \int \pi(x_i, e_i \mid \theta) r(\theta) d\theta$, where $r(\theta)$ is a prior probability distribution. Since the above inequality holds for every value of θ and $r(\theta) \geq 0$, we can multiply both sides by $r(\theta)$, and integrate. Notice that the right hand side is not a function of θ and it remains unchanged because $\int r(\theta) d\theta = 1$:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i \mid e_i} \parallel P_{X_i \mid e_i} \right) \right] \leq \sup_{\theta \in \Theta} \sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{R(x_i \mid e_i)}{P(x_i \mid e_i)} \cdot$$

Now from Inequality 9, for *any* $P \in \mathcal{P}$ and $R \in \mathcal{P}$, it follows that

$$\mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i \mid e_i} \parallel P_{X_i \mid e_i} \right) \right] \leq \sup_{\theta \in \Theta} \mathbb{D}_{\mathcal{X}}[P \mid \theta],$$

which proves the lemma. □

Theorem 2. *Suppose $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is a parametric model, and let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ be a query set in $\mathcal{M}_\Theta$. If for every value of θ there is a distribution $R \in \mathcal{P}$ such that $R(s) = \pi(s \mid \theta)$ for every $s \in \mathcal{S}$, then for any $P \in \mathcal{P}$ we have:*

$$\mathbb{D}_{\mathcal{X}}[P] = \sup_{\theta \in \Theta} \mathbb{D}_{\mathcal{X}}[P \mid \theta] \cdot \tag{10}$$

Proof. For an arbitrary distribution $P \in \mathcal{P}$, we let

$$\theta^* = \arg\sup_{\theta \in \Theta} \mathbb{D}_{\mathcal{X}}[P \mid \theta] \cdot$$

In $\mathcal{M}_\Theta$ there should be a distribution $R \in \mathcal{P}$ that equals $\pi(s \mid \theta^*)$ for every $s \in \mathcal{S}$. Obviously, R and $\pi(\cdot \mid \theta^*)$ define the same distribution for any random variable, and we have

$$\begin{aligned}
\mathbb{E}_R\left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right)\right] &= \mathbb{E}_\pi\left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta^*} \parallel P_{X_i|e_i}\right) \,\middle|\, \theta^*\right] \\
&= \sup_{\theta \in \Theta} \mathbb{D}_{\mathcal{X}}[P \mid \theta] \cdot
\end{aligned}$$

On the other hand, clearly

$$\sup_{Q \in \mathcal{P}} \mathbb{E}_Q\left[\sum_i \mathbb{D}\left(Q_{X_i|e_i} \parallel P_{X_i|e_i}\right)\right] \geq \mathbb{E}_R\left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right)\right] \cdot$$

Notice that the left hand side is $\mathbb{D}_{\mathcal{X}}[P]$, and hence

$$\mathbb{D}_{\mathcal{X}}[P] \geq \sup_{\theta \in \Theta} \mathbb{D}_{\mathcal{X}}[P \mid \theta] \cdot$$

This inequality and Lemma 1 prove the theorem. □

Proposition. A generic parametric model satisfies the conditions of Theorem 2.

Theorem 3. *Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a generic parametric model and $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ be a query set. We define*

$$f(P_{\mathcal{X}}, r) = \int \left(\sum_i \sum_{x_i, e_i} \pi(x_i \mid e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{p_{x_i|e_i}}\right) r(\theta) d\theta,$$

where r is a probability distribution, and $P_{\mathcal{X}}$ is a vector that includes an entry $p_{x_i|e_i} \geq 0$ for every i, x_i, e_i, such that $\sum_{x_i} p_{x_i|e_i} = 1$.

f is a concave-convex function and has a saddle point: $(r^*, P^*_{\mathcal{X}})$. Furthermore, the distribution $P(X_i = x_i \mid E_i = e_i) = p^*_{x_i|e_i}$ is a central distribution for $\mathcal{X}$. If $(P^*_{\mathcal{X}}, r^*)$ is a saddle point of f, and there is a prior distribution p such that $p^*_{x_i|e_i} = \int \pi(x_i \mid e_i, \theta)p(\theta)d\theta$, then the distribution $P(X_i = x_i \mid E_i = e_i) = p^*_{x_i|e_i}$ is a central distribution for $\mathcal{X}$.

Proof. We define

$$f(P_{\mathcal{X}}, r) = \int \left(\sum_i \sum_{x_i, e_i} \pi(x_i \mid e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{p_{x_i|e_i}} \right) r(\theta) d\theta,$$

where r is a real function of the model parameter θ, and $P_{\mathcal{X}}$ is a vector of real numbers with an entry $p_{x_i|e_i}$ for every i, x_i, e_i.

Let $(P^*_{\mathcal{X}}, r^*)$ be a simultaneous solution to the following two optimization problems:

$$\begin{array}{lll} \text{minimize} & f(P_{\mathcal{X}}, r^*) & \text{w.r.t. } P_{\mathcal{X}} \\ \text{subject to} & \displaystyle\sum_{x_i} p_{x_i|e_i} = 1, & \text{for all } i, e_i \end{array}$$

$$\begin{array}{lll} \text{maximize} & f(P^*_{\mathcal{X}}, r) & \text{w.r.t. } r \\ \text{subject to} & r(\theta) \geq 0, & \text{for all } \theta \\ & \displaystyle\int r(\theta) d\theta = 1 & \end{array}$$

Clearly, if $(P^*_{\mathcal{X}}, r^*)$ exists, it will be a saddle point for f.

Both of these optimization problems are convex and differentiable [4]. All the constraint functions are affine, and therefore are convex and differentiable. Note that for the constraint $r(\theta) \geq 0$, we can define an infinite number of affine constraint functions of the form $\int \delta(\theta - t) r(t) dt$, where δ is the Dirac delta function.

Additionally, f is differentiable with respect to both r and $P_{\mathcal{X}}$. For any fixed $P_{\mathcal{X}}$, $f(P_{\mathcal{X}}, \cdot)$ is an affine function and is concave. For a fixed r^*, we can write

$$f(P_{\mathcal{X}}, r^*) = c - \sum_i \sum_{x_i, e_i} R^*(x_i, e_i) \ln p_{x_i|e_i}, \tag{11}$$

where c and $R^*(x_i, e_i) = \int \pi(x_i, e_i \mid \theta) r^*(\theta) d\theta$ are not functions of $P_{\mathcal{X}}$. Since r^* is a solution to the second optimization problem, it is feasible and satisfies all the constraints. Therefore, $R^*(x_i, e_i) \geq 0$ for any i, e_i, and $f(\cdot, r^*)$ is the negative of a linear combination of concave $\ln p_{x_i|e_i}$ functions with nonnegative coefficients and is a convex function.

Since both optimization problems are convex and differentiable, any point $(P^*_{\mathcal{X}}, r^*)$ that simultaneously satisfies the KKT conditions [4] for both problems is a saddle point for f. The following KKT conditions can be obtained for every θ, i, x_i, and e_i:

$$\int \pi(x_i \mid e_i, \theta) r^*(\theta) d\theta = \lambda(e_i) p^*_{x_i|e_i}, \tag{12}$$

$$\sum_{x_i} p^*_{x_i|e_i} = 1, \tag{13}$$

$$\sum_i \sum_{x_i, e_i} \pi(x_i \mid e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{p^*_{x_i|e_i}} - \mu(\theta) = \lambda,$$

$$\int r^*(\theta) d\theta = 1, \tag{14}$$

$$r^*(\theta) \geq 0 \tag{15}$$

$$\mu(\theta) \geq 0,$$

$$\mu(\theta) r^*(\theta) = 0 \cdot \tag{16}$$

By summing Equation 12 over x_i and using Equations 14 and 13, we obtain $\lambda(e_i) = 1$.

Note that we can impose more restrictive conditions and still derive a set of sufficient conditions for a saddle point. Hence, we can replace Inequality 15 with the more restrictive $r^*(\theta) > 0$. Then, Equation 16 implies $\mu(\theta) = 0$, and the KKT conditions are simplified to

$$\sum_i \sum_{x_i, e_i} \pi(x_i \mid e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{R^*(x_i \mid e_i)} = \lambda, \tag{17}$$

$$R^*(x_i \mid e_i) = \int \pi(x_i \mid e_i, \theta) r^*(\theta) d\theta,$$

$$r^*(\theta) > 0 \cdot$$

Let $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model. If a probability measure $R^* \in \mathcal{P}$ which is obtained by a positive prior, satisfies Equation 17, then it corresponds to a saddle point of f Suppose $C_{\mathcal{X}}$ is a central distribution for $\mathcal{X}$. Since $R^* \in \mathcal{P}$, we have

$$\mathbb{D}_{\mathcal{X}} [C_{\mathcal{X}}] \leq \mathbb{D}_{\mathcal{X}} [R^*] \cdot$$

On the other hand, $(R^*_{\mathcal{X}}, r^*)$ is a saddle point and we have

$$f(R^*_{\mathcal{X}}, r^*) \leq f(C_{\mathcal{X}}, r^*) \tag{18}$$

r^* is a distribution. Therefore,

$$f(C_{\mathcal{X}}, r^*) = \int \mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}} \mid \theta\right] r^*(\theta) d\theta \leq \sup_{\theta \in \Theta} \mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}} \mid \theta\right] = \mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}}\right]$$

On the other hand, Equation 17 implies

$$\mathbb{D}_{\mathcal{X}}\left[R^*\right] = f(R^*_{\mathcal{X}}, r^*) = \lambda$$

By inequality 18 we get

$$\mathbb{D}_{\mathcal{X}}\left[R^*\right] \leq \mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}}\right]$$

Thus, we proved

$$\mathbb{D}_{\mathcal{X}}\left[R^*\right] = \mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}}\right],$$

which completes the proof of the theorem. □

Proof. r^* is a distribution and $f(C_{\mathcal{X}}, r^*)$ is a weighted average of $\mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}} \mid \theta\right]$ for different values of θ. Therefore,

$$f(C_{\mathcal{X}}, r^*) \leq \mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}}\right]$$

where R is a probability measure in $\mathcal{M}_{\Theta}$ that is obtained by a positive prior $r^*(\theta) > 0$:

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of concave functions with non-positive we have that ensures $f(\cdot, r^*)$ is convex. Note that is a solution we use kkt [4] First, we show that f is a concave-convex function.

For any fixed $P_{\mathcal{X}}$, $f(P_{\mathcal{X}}, \cdot)$ is a linear function. Note that if a function is concave (convex) on a set, it is also concave (convex) on any convex subset of that set. The domain of $f(P_{\mathcal{X}}, \cdot)$ is the set of all real functions r such that $\int r(\theta) d\theta = 1$. We can easily verify that this set is a convex set, and therefore $f(P_{\mathcal{X}}, \cdot)$ is concave (and also convex) on the defined domain.

On the other hand, for a fixed r, if we let $Q(x_i, e_i) = \int \pi(x_i, e_i \mid \theta) r(\theta) d\theta$ we have

$$f(\cdot, r) = c - \sum_{i} \sum_{x_i, e_i} Q(x_i, e_i) \ln p_{x_i|e_i}, \tag{19}$$

where c and $Q(x_i, e_i)$ are constants. We know that a function of the form $f(x_1, x_2, \ldots) = \sum_i a_i \ln x_i$ is concave for $a_i \geq 0$. Consequently, $f(\cdot, r)$ will be convex if the domain of valid $P_{\mathcal{X}}$-vectors is a compact convex set. For a valid $P_{\mathcal{X}}$ we must have $p_{x_i|e_i} \in [0, 1]$ and $\sum_{x_i} p_{x_i|e_i} = 1$. We can easily verify that if $P_{\mathcal{X}}$ and $P'_{\mathcal{X}}$ satisfy the required conditions, $\alpha P_{\mathcal{X}} + (1-\alpha) P'_{\mathcal{X}}$, where $\alpha \in [0, 1]$ also satisfies these conditions. Therefore, $f(\cdot, r)$ is a convex function.

Since f is a differentiable concave-convex function, any point $(P^*_{\mathcal{X}}, r^*)$ that is simultaneously a stationary point of the two Lagrangian functions

$$\mathcal{L}\left[r(\theta), \lambda\right] = \int \left(\sum_i \sum_{x_i, e_i} \pi(x_i \mid e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{p_{x_i|e_i}} \right) r(\theta) d\theta \\ - \lambda \left(\int r(\theta) d\theta - 1 \right),$$

and

$$\mathcal{L}\left[P_{\mathcal{X}}, \lambda_{e_i}\right] = \int \left(\sum_i \sum_{x_i, e_i} \pi(x_i \mid e_i, \theta) \ln \frac{\pi(x_i \mid e_i, \theta)}{p_{x_i|e_i}} \right) r(\theta) d\theta \\ + \sum_i \sum_{e_i} \left(\lambda_{e_i} \left(\sum_{x_i} p_{x_i|e_i} - 1 \right) \right).$$

is a saddle point of f. [1]

By setting the partial derivatives of the Lagrangian functions to zero, we obtain the following system of equations for every θ, i, x_i, and e_i:

Any solution $(P^*_{\mathcal{X}}, r^*)$ to this equation system is a saddle points of f.

$$-\int \frac{\pi(x_i \mid e_i, \theta)}{p_{x_i|e_i}} r(\theta) d\theta + \lambda(e_i)$$

From Equation 21, it follows that the domain of the $P_{X_i|E_i}$-vectors is also a compact convex set. If we calculate the Hessian matrix of $f(\cdot, r)$, we observe that the matrix is diagonal, and each element of the main diagonal is $\frac{R(x_i|e_i)}{P(x_i|e_i)^2}$. Thus, the hessian matrix of $f(\cdot, r)$ is positive semi-definite and $f(\cdot, r)$ is a convex function.

Let $(P^*_{\mathcal{X}}, r^*)$ be the saddle point of f, and $C_{\mathcal{X}}$ be the central distribution for $\mathcal{X}$. For a saddle point we have

$$f(P^*_{\mathcal{X}}, r^*) = \min_{P_{\mathcal{X}}} f(P_{\mathcal{X}}, r^*),$$

If $P_{\mathcal{X}}$ represents a valid distribution for each $X_i|E_i$ that can be obtained by a prior distribution, we have

$$\mathbb{D}_{\mathcal{X}}\left[P^*_{\mathcal{X}}\right] = f(P^*_{\mathcal{X}}, r^*)$$

[1] I need reference for this! This seems similar to the case where $f'(x*) = 0$ for a differentiable convex function implies that $x*$ is a global minimum. However, I'm still looking for a good reference for this.

That impleas

$$\mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}}\right] \leq f(P^*_{\mathcal{X}}, r^*)$$

$$\begin{aligned} f(P^*_{\mathcal{X}}, r^*) &= \max_{r} f(P^*_{\mathcal{X}}, r) \\ &= \min_{P_{\mathcal{X}}} f(P_{\mathcal{X}}, r^*)\cdot \end{aligned}$$

This implies

$$\mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}}\right] \geq f(C_{\mathcal{X}}, r^*) \geq f(P^*_{\mathcal{X}}, r^*)$$

and $C_{\mathcal{X}}$ be the central distribution for $\mathcal{X}$.

sdada

$$\int Fp(\theta)d\theta$$

since

$$f(P^*_{\mathcal{X}}, r^*) = \mathbb{D}_{\mathcal{X}}\left[P_{\mathcal{X}}\right]$$

□

Theorem 2 implies that in a generic parametric model, we can obtain the central probability measure by:

$$C_{\mathcal{X}} = \arg\min_{P \in \mathcal{P}} \sup_{\theta \in \Theta} \mathbb{D}_{\mathcal{X}}\left[P \mid \theta\right]\cdot$$

This is a challenging minimax problem that cannot be easily solved. However, by using the minimax theorem we can convert it into a more convenient maximization problem which can be solved by variational methods.

Theorem 4 (Minimax [5])**.** *Let $\mathcal{X} \subset \mathbb{R}^n$ and $\mathcal{Y} \subset \mathbb{R}^m$ be compact convex sets. If $f : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ is a continuous function that is concave-convex, i.e.*

- *$f(\cdot, y) : \mathcal{X} \to \mathbb{R}$ is concave for fixed y, and*
- *$f(x, \cdot) : \mathcal{Y} \to \mathbb{R}$ is convex for fixed x.*

Then we have that

$$\max_{x \in \mathcal{X}} \min_{y \in \mathcal{Y}} f(x, y) = \min_{y \in \mathcal{Y}} \max_{x \in \mathcal{X}} f(x, y)\cdot \tag{20}$$

A vector in $\mathbb{R}^n$ shares many properties with a real function. We sometimes assume that a real function is a real vector, and vice versa. In the next theorem we will use the minimax theorem, assuming x is a real function.

Theorem 5. *Consider a generic parametric model $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$, and let $\mathcal{X} = \{(X_1, E_1), (X_2, E_2), \ldots\}$ be a query set in $\mathcal{M}_\Theta$. Assume that for every $Q_1, Q_2 \in \mathcal{P}$ and any real number $0 \leq \alpha \leq 1$, there is a distribution $Q^* \in \mathcal{P}$, such that for every i, x_i, e_i we have*

$$Q^*(x_i \mid e_i) = \alpha Q_1(x_i \mid e_i) + (1-\alpha) Q_2(x_i \mid e_i)\cdot \tag{21}$$

Then, we have

$$\min_{P\in\mathcal{P}} \sup_{\theta\in\Theta} \mathbb{D}_{\mathcal{X}}\left[P \mid \theta\right] = \max_{P\in\mathcal{P}} \int \mathbb{D}_{\mathcal{X}}\left[P \mid \theta\right] p(\theta) d\theta\cdot \tag{22}$$

Proof. We prove this theorem using the minimax theorem. To do so, we define a function as follows:

$$f(P_{\mathcal{X}}, r) = \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel P_{X_i|e_i}\right) \,\middle|\, \theta \right] r(\theta) d\theta\cdot$$

Here, $r(\theta)$ is an arbitrary probability distribution, and $P_{X_i|E_i}$ is a vector that includes all values of $P(X_i = x_i \mid E_i = e_i)$ for every i, x_i, e_i.

Since $r(\theta)$ can be any distribution, maximizing with respect to r is similar to maximizing with respect to θ, and we have

$$\min_{P\in\mathcal{P}} \sup_{\theta\in\Theta} \mathbb{D}_{\mathcal{X}}\left[P \mid \theta\right] = \min_{P\in\mathcal{P}} \max_r f(P_{\mathcal{X}}, r)\cdot$$

Hence, we can use Theorem 4

$$\min_{P\in\mathcal{P}} \max_r f(P_{\mathcal{X}}, r) = \max_r \min_{P\in\mathcal{P}} f(P_{\mathcal{X}}, r)\cdot \tag{23}$$

From Equation 19 we can see that for an arbitrary fixed r we have

$$\min_{P\in\mathcal{P}} f(P_{\mathcal{X}}, r) = \max_{P\in\mathcal{P}} \sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln P(x_i \mid e_i)\cdot$$

We know that the Kullback–Leibler divergence, and therefore its expected value, is non-negative. Consequently:

$$\mathbb{E}_R \left[\sum_i \mathbb{D}\left(R_{X_i|e_i} \parallel P_{X_i|e_i}\right) \right] \geq 0\cdot$$

By simple algebra we get

$$\sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln P(x_i \mid e_i) \leq \sum_i \sum_{x_i, e_i} R(x_i, e_i) \ln R(x_i \mid e_i)\cdot$$

Recall that $\mathcal{M}_\Theta$ is a generic parametric model. That implies $R \in \mathcal{P}$, and we can reach this upper bound by letting $P = R$. Hence, for a fixed r

$$\min_{P\in\mathcal{P}} f(P_\mathcal{X}, r) = \int \mathbb{E}_\pi \left[\sum_i \mathbb{D}\left(\pi_{X_i|e_i,\theta} \parallel R_{X_i|e_i}\right) \,\middle|\, \theta \right] r(\theta)d\theta\cdot$$

Substituting this into Equation 23 completes the proof of the theorem. $\square$

Let us now use this theorem for our dice example and rewrite Equation 8:

$$\begin{aligned} C_\mathcal{X} &= \arg\max_{P\in\mathcal{P}} \int \sum_{x_1,\ldots,x_n} \pi(x_1,\ldots,x_n \mid \theta) \ln \frac{\pi(x_1,\ldots,x_n \mid \theta)}{P(x_1,\ldots,x_n)} p(\theta)d\theta \\ &= \arg\max_{P\in\mathcal{P}} \int \sum_{x_1,\ldots,x_n} p(x_1,\ldots,x_n,\theta) \ln \frac{p(\theta \mid x_1,\ldots,x_n)}{p(\theta)} d\theta\cdot \end{aligned}$$

This means that the central distribution for our dice example, when we consider a query set equivalent to the sample space, is the probability measure that is obtained by using the Reference prior [3] for the multinomial parameters.

Theorem 6. *Suppose the conditions of Theorem 5 hold and the central distribution for $\mathcal{X}$ exists and can be obtained by*

$$C_\mathcal{X} = \arg\max_{P\in\mathcal{P}} \int \mathbb{D}_\mathcal{X}\left[P \mid \theta\right] p(\theta)d\theta\cdot$$

in $\mathcal{M}_\Theta$, if any prior that is negative for some values of θ, induces a negative distribution for a variable in $\mathcal{X}$, then the central distribution for $\mathcal{X}$ satisfies:

$$\mathbb{D}_\mathcal{X}\left[C_\mathcal{X} \mid \theta\right] = \lambda, \tag{24}$$

where λ is a constant.

Proof. By the assumptions of the theorem, the central distribution for $\mathcal{X}$ is a solution to the problem of maximizing

$$\mathcal{F}\left[p(\theta)\right] = \int \sum_i \sum_{x_i,e_i} p(x_i,e_i,\theta) \ln \frac{\pi(x_i \mid e_i,\theta)}{P(x_i \mid e_i)} d\theta,$$

with respect to different choices of the prior distribution $p(\theta)$, subject to the constraint $\int p(\theta)d\theta = 1$. We do not need to consider the constraint $p(\theta) \geq 0$, since the objective function $\mathcal{F}$ contains logarithm terms that ensure all $P(x_i \mid e_i)$ will be positive in any solution. By assumption, this implies $p(\theta) \geq 0$.

Applying the Lagrange multiplier technique, the solution is a stationary point of the Lagrangian functional:

$$\mathcal{L}\left[p(\theta),\lambda\right] = \int \sum_i \sum_{x_i,e_i} p(x_i,e_i,\theta) \ln \frac{\pi(x_i \mid e_i,\theta)}{P(x_i \mid e_i)} d\theta - \lambda \left(\int p(\theta) d\theta - 1 \right) \cdot$$

Let us denote the prior distribution corresponding to the central distribution $C_{\mathcal{X}}$ by $c_{\mathcal{X}}(\theta)$. Let $p(\theta) = c_{\mathcal{X}}(\theta) + \epsilon\eta(\theta)$ be a variation of $c_{\mathcal{X}}(\theta)$, where $\eta(\theta)$ is an arbitrary continuously differentiable function with $\eta(0) = \eta(1) = 0$, and ϵ is a small constant.

Since $c_{\mathcal{X}}(\theta)$ is an extremizing function for $\mathcal{F}$, the lagrangian $\mathcal{L}\left[\epsilon,\lambda\right]$ has a stationary point where $\epsilon = 0$, and we have

$$\left. \frac{\partial \mathcal{L}\left[\epsilon,\lambda\right]}{\partial \epsilon} \right|_{\epsilon=0} = 0 \cdot$$

Note that

$$\begin{aligned} \frac{dp(\theta)}{d\epsilon} &= \eta(\theta), \\ \frac{dP(x_i,e_i)}{d\epsilon} &= \int \pi(x_i,e_i \mid \theta)\eta(\theta) d\theta, \\ \frac{dP(e_i)}{d\epsilon} &= \int \pi(e_i \mid \theta)\eta(\theta) d\theta \cdot \end{aligned}$$

Using the chain rule of calculus, we calculate the partial derivative of $\mathcal{L}\left[\epsilon,\lambda\right]$ relative to ϵ:

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial \epsilon} = &\int \left(\sum_i \sum_{x_i,e_i} \pi(x_i,e_i \mid \theta) \ln \frac{\pi\left(x_i \mid e_i,\theta\right)}{P\left(x_i \mid e_i\right)} \right) \eta(\theta) d\theta \\ &- \int \left(\sum_i \sum_{x_i,e_i} \pi(x_i,e_i \mid \theta) \frac{\int \pi(x_i,e_i \mid \theta)\eta(\theta) d\theta}{P(x_i,e_i)} \right) p(\theta) d\theta \\ &+ \int \left(\sum_i \sum_{e_i} \pi(e_i \mid \theta) \frac{\int \pi(e_i \mid \theta)\eta(\theta) d\theta}{P(e_i)} \right) p(\theta) d\theta \\ &- \int \lambda\eta(\theta) d\theta \end{aligned}$$

Since $P(x_i,e_i) = \int \pi(x_i,e_i \mid \theta)p(\theta)d\theta$ and $P(e_i) = \int \pi(e_i \mid \theta)p(\theta)d\theta$, we get

$$\frac{\partial \mathcal{L}}{\partial \epsilon} = \int \left(-\lambda + \sum_i \sum_{x_i,e_i} \pi(x_i,e_i \mid \theta) \ln \frac{\pi\left(x_i \mid e_i,\theta\right)}{P\left(x_i \mid e_i\right)} \right) \eta(\theta) d\theta \cdot$$

We let $\frac{\partial \mathcal{L}}{\partial \epsilon}\Big|_{\epsilon=0} = 0$. Since $\eta(\theta)$ is an arbitrary function, by applying the fundamental lemma of the calculus of variations and letting $\epsilon = 0$, for every value of θ we have

$$\sum_i \sum_{x_i, e_i} \pi(x_i, e_i \mid \theta) \ln \frac{\pi\left(x_i \mid e_i, \theta\right)}{C_{\mathcal{X}}\left(x_i \mid e_i\right)} = \lambda \cdot \qquad \square$$

This theorem gives a necessary condition for the central probability measure in the general case.

Theorem 7. *Assume $\mathcal{M}_\Theta \equiv (\mathcal{S}, \mathcal{P})$ is a generic parametric model, and $\mathcal{X} = \{X, X_1, X_2, \ldots, X_k\}$ is a query set in $\mathcal{M}_\Theta$. Let Ω_i, for every i, be a function of the parameter Θ that fully parametrizes X_i, That is, for $\omega_i = \Omega_i(\theta)$ we have:*

$$\pi(x_i \mid \theta) = \pi\left(x_i \mid \omega_i\right) \cdot \tag{25}$$

Also, let $\Omega(\theta) = (\omega_1, \omega_2, \ldots, \omega_k) = \omega$.

We define a map T on the space of unnormalized probability measures of ω as follows:

$$T\left[\rho\right](\omega) = \left(\frac{\varphi_\rho(\omega)}{\rho(\omega)} \prod_i \frac{\varphi_\rho(\omega_i)}{\rho(\omega_i)} \right)^{\frac{1}{k+1}} \rho(\omega), \tag{26}$$

where ρ is an unnormalized probability measure and:

$$\begin{aligned}
\varphi_\rho(\omega_i) &= \exp\left\{ \sum_{x_i} \pi(x_i \mid \omega_i) \ln \rho(\omega_i \mid x_i) \right\}, \\
\varphi_\rho(\omega) &= \int_{\{\theta : \Omega(\theta) = \omega\}} \varphi_\rho(\theta) d\theta, \\
\varphi_\rho(\theta) &= \exp\left\{ \sum_x \pi(x \mid \theta) \ln \rho(\theta \mid x) \right\} \cdot
\end{aligned}$$

If $c_{\mathcal{X}}$ denotes the probability mass function corresponding to the central probability measure for $\mathcal{X}$ in $\mathcal{M}_\Theta$, then $\mu c_{\mathcal{X}}$ is a fixed point for T, where

$$\mu = \exp\left\{ \frac{1}{k+1} \int \mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}} \mid \theta\right] c_{\mathcal{X}}(\theta) d\theta \right\} \cdot$$

Moreover, we have

$$c_{\mathcal{X}}(\theta \mid \omega) = \frac{\varphi_{c_{\mathcal{X}}}(\theta)}{\varphi_{c_{\mathcal{X}}}(\omega)} \cdot \tag{27}$$

Proof. Let $Q^*(y) = \alpha Q_1(y) + (1-\alpha)Q_2(y)$, for some $Q_1, Q_2 \in \mathcal{P}$, $0 \leq \alpha \leq 1$ and an arbitrary random variable Y. $\mathcal{M}_\Theta$ is a parametric model. Therefore:

$$\begin{aligned} Q^*(y) &= \alpha \int \pi(y \mid \theta) q_1(\theta) d\theta + (1-\alpha) \int \pi(y \mid \theta) q_2(\theta) d\theta \\ &= \int \pi(y \mid \theta)(\alpha q_1(\theta) + (1-\alpha) q_2(\theta)) d\theta \cdot \end{aligned}$$

$\alpha q_1(\theta) + (1-\alpha)q_2(\theta)$ is a valid probability distribution and $\mathcal{M}_\Theta$ is a generic parametric model. That implies $Q^* \in \mathcal{P}$. This proves that the set of distributions of X and each X_i is a convex set.

Thus, the conditions of Theorem 6 holds and the central distribution of $\mathcal{X}$ satisfies

$$\mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}} \mid \theta\right] = \lambda \cdot$$

Moreover, from Equation 25 and Bayes' theorem it follows

$$\begin{aligned} \mathbb{D}_{\mathcal{X}}\left[C_{\mathcal{X}} \mid \theta\right] &= \sum_x \pi(x \mid \theta) \ln \frac{\pi(x \mid \theta)}{C_{\mathcal{X}}(x)} + \sum_i \sum_{x_i} \pi(x_i \mid \theta) \ln \frac{\pi(x_i \mid \theta)}{C_{\mathcal{X}}(x_i)} \\ &= \sum_x \pi(x \mid \theta) \ln \frac{c_{\mathcal{X}}(\theta \mid x)}{c_{\mathcal{X}}(\theta)} + \sum_i \sum_{x_i} \pi(x_i \mid \omega_i) \ln \frac{c_{\mathcal{X}}(\omega_i \mid x_i)}{c_{\mathcal{X}}(\omega_i)} \\ &= \ln \frac{\varphi_{c_{\mathcal{X}}}(\theta)}{c_{\mathcal{X}}(\theta)} + \sum_i \ln \frac{\varphi_{c_{\mathcal{X}}}(\omega_i)}{c_{\mathcal{X}}(\omega_i)} \cdot \end{aligned}$$

That implies

$$e^\lambda = \frac{\varphi_{c_{\mathcal{X}}}(\theta)}{c_{\mathcal{X}}(\theta)} \prod_i \frac{\varphi_{c_{\mathcal{X}}}(\omega_i)}{c_{\mathcal{X}}(\omega_i)} \cdot \tag{28}$$

For $\Omega(\theta) = \omega$, we have $c_{\mathcal{X}}(\theta, \omega) = c_{\mathcal{X}}(\theta)$. Hence,

$$c_{\mathcal{X}}(\theta) = c_{\mathcal{X}}(\theta \mid \omega) c_{\mathcal{X}}(\omega) \cdot$$

We substitute this equation into Equation 28:

$$c_{\mathcal{X}}(\theta \mid \omega) e^\lambda = \frac{\varphi_{c_{\mathcal{X}}}(\theta)}{c_{\mathcal{X}}(\omega)} \prod_i \frac{\varphi_{c_{\mathcal{X}}}(\omega_i)}{c_{\mathcal{X}}(\omega_i)} \cdot \tag{29}$$

Since this equation holds for any value of θ, we can integrate both sides with respect to θ:

$$\int_{\{\theta : \Omega(\theta) = \omega\}} c_{\mathcal{X}}(\theta \mid \omega) e^\lambda d\theta = \int_{\{\theta : \Omega(\theta) = \omega\}} \frac{\varphi_{c_{\mathcal{X}}}(\theta)}{c_{\mathcal{X}}(\omega)} \prod_i \frac{\varphi_{c_{\mathcal{X}}}(\omega_i)}{c_{\mathcal{X}}(\omega_i)} d\theta \cdot$$

Therefore,

$$e^{\lambda} = \frac{\varphi_{c_{\mathcal{X}}}(\omega)}{c_{\mathcal{X}}(\omega)} \prod_i \frac{\varphi_{c_{\mathcal{X}}}(\omega_i)}{c_{\mathcal{X}}(\omega_i)} \cdot \tag{30}$$

Now we evaluate T at $\mu c_{\mathcal{X}}$:

$$T\left[\mu c_{\mathcal{X}}\right](\omega) = \left(\frac{\varphi_{\mu c_{\mathcal{X}}}(\omega)}{\mu c_{\mathcal{X}}(\omega)} \prod_i \frac{\varphi_{\mu c_{\mathcal{X}}}(\omega_i)}{\mu c_{\mathcal{X}}(\omega_i)}\right)^{\frac{1}{k+1}} \mu c_{\mathcal{X}}(\omega) = e^{\frac{\lambda}{k+1}} c_{\mathcal{X}}(\omega) \cdot$$

Clearly for $\mu = e^{\frac{\lambda}{k+1}}$, $\mu c_{\mathcal{X}}$ is a fixed point for T.

In addition, we can obtain a closed form for $c_{\mathcal{X}}(\theta \mid \omega)$ by dividing Equation 29 by Equation 30:

$$c_{\mathcal{X}}(\theta \mid \omega) = \frac{\varphi_{c_{\mathcal{X}}}(\theta)}{\varphi_{c_{\mathcal{X}}}(\omega)},$$

which completes the proof. □

Using Equation 26, we can construct a sequence of unnormalized measures by

$$\rho_{n+1} = T\left[\rho_n\right] \cdot$$

This sequence can be used in an iterative algorithm to find an estimation of the central probability measure $c_{\mathcal{X}}(\omega)$. Once $c_{\mathcal{X}}(\omega)$ is determined, Equation 27 can be used to obtain $c_{\mathcal{X}}(\theta)$.

The key property of Equation 26 is that it allows us to compute $c_{\mathcal{X}}(\theta)$ for specific regions of the parameter space without needing to scan the entire space, which is typically large and intractable. Additionally, it defines a mapping based on ω and its marginals ω_i, rather than θ. Notably, the space of ω is significantly smaller than that of θ.

Definition 2.7 (Posterior Potential)**.** Let $\mathcal{M}_{\Theta} \equiv (\mathcal{S}, \mathcal{P})$ be a parametric model and $\mathcal{X}$ be a query set. Let $X \in \mathcal{X}$, and Ω be a function of the parameter Θ that fully parametrizes X. We define the *posterior potential* of X for ω, denoted by $\varphi_X(\omega)$, as follows:

$$\varphi_X(\omega) = \exp\left\{\sum_x \pi(x \mid \omega) \ln c_{\mathcal{X}}(\omega \mid x)\right\} \cdot$$

When X is clear from the context we will omit X from the notation and simply write $\varphi(\omega)$.

In Equation 26, if we replace φ_ρ with $\varphi_{c_\mathcal{X}}$, that could improve the convergence properties considerably. In a model that the posterior of X converges, we can find φ_X by using any well behaved prior instead of $c_\mathcal{X}(\theta)$. We typically will use a conjugate prior to simplify the derivation of the posterior potential. Using a conjugate prior, the posterior potential takes a form similar to the entropy of the likelihood distribution.

Theorem 8. *Let T be the map defined in theorem 7. The sequence*

$$\rho_{n+1} = T\left[\rho_n\right],$$

converges to the fixed point of T for any arbitrary initial unnormalized probability measure ρ_1.

Proof. We prove the theorem by using Banach fixed point theorem [1]. Let ρ and η be two unnormalized probability measures of a random variable ω We define the distance of ρ and η as follows:

$$d_\Omega\left(\rho, \eta\right) = \int \left| \ln \frac{\rho(\omega)}{\eta(\omega)} \right| d\omega\cdot \tag{31}$$

Obviously, if $\rho = \eta$, then $d_\Omega\left(\rho, \eta\right) = 0$. Conversely, if $d_\Omega\left(\rho, \eta\right) = 0$, that implies $\rho = \eta$ because $\int |f(\omega)|\, d\omega = 0$ if and only if $f(\omega) = 0$. Moreover, d_Ω is symmetric $d_\Omega\left(\rho, \eta\right) = d_\Omega\left(\eta, \rho\right)$, and satisfies the triangle inequality:

$$\begin{aligned} d_\Omega\left(\rho, \phi\right) + d_\Omega\left(\phi, \eta\right) &= \int \left| \ln \frac{\rho(\omega)}{\phi(\omega)} \right| d\omega + \int \left| \ln \frac{\phi(\omega)}{\eta(\omega)} \right| d\omega \\ &= \int \left(\left| \ln \frac{\rho(\omega)}{\phi(\omega)} \right| + \left| \ln \frac{\phi(\omega)}{\eta(\omega)} \right| \right) d\omega \\ &\geq \int \left| \ln \frac{\rho(\omega)}{\phi(\omega)} + \ln \frac{\phi(\omega)}{\eta(\omega)} \right| d\omega \\ &= d_\Omega\left(\rho, \eta\right)\cdot \end{aligned}$$

Thus, d_Ω is a metric.

Now, we show that T is a contraction.

$$\begin{aligned}
d_\Omega\left(T\left[\rho\right], T\left[\eta\right]\right) &= \int \left| \ln \frac{T\left[\rho\right](\omega)}{T\left[\eta\right](\omega)} \right| d\omega \\
&= \int \left| \ln \left(\frac{\rho(\omega)}{\eta(\omega)} \left(\frac{\eta(\omega)}{\rho(\omega)} \prod_i \frac{\eta(\omega_i)}{\rho(\omega_i)} \right)^{\frac{1}{k+1}} \right) \right| d\omega \\
&= \frac{1}{k+1} \int \left| \ln \left(\prod_i \frac{p_\rho(\omega \mid \omega_i)}{p_\eta(\omega \mid \omega_i)} \right) \right| d\omega \\
&\leq \frac{1}{k+1} \sum_{i=1}^{k} \int \left| \ln \frac{p_\rho(\omega \mid \omega_i)}{p_\eta(\omega \mid \omega_i)} \right| d\omega \\
&\leq^? \frac{k}{k+1} d_\Omega\left(\rho, \eta\right)
\end{aligned}$$

Unfortunately, this proof is not complete. □

2.3 Multinomial Model

In this section, we examine the derivation of the central probability measure for multinomial models. Multinomial models play a key role in statistical learning. By de Finetti's theorem, any exchangeable sequence of discrete random variables can be represented using a multinomial model. Exchangeable sequences are fundamental to statistical learning, as they offer a framework for modeling data with independence and order invariance.

A multinomial model is a parametric model (Definition 1.2) in which the likelihood function follows a multinomial distribution. The configuration of the model is determined by two variables: n and d. Each element of the sample space $s \in \mathcal{S}$ can be represent by a d dimensional random vector

$$X = (x_1, x_2, \ldots, x_d),$$

where each x_i is a nonnegative integer such that $\sum_{i=1}^d x_i = n$. The model parameter is a nonnegative d dimensional real vector

$$\theta = (\theta_1, \theta_2, \ldots, \theta_d),$$

where $\sum_{i=1}^d \theta_i = 1$. The likelihood distributing $\pi(X \mid \theta)$ is a multinomial distribution

$$\pi(x \mid \theta) = \frac{n!}{\prod_{i=1}^d x_i!} \prod_{i=1}^d \theta_i^{x_i}.$$

As n increases, the posterior distribution of θ converges to the same distribution regardless of the prior. Therefore, when n is sufficiently large, we can use a uniform prior to obtain the posterior potential of the multinomial parameter. The uniform prior is conjugate to the multinomial distribution, and the resulting posterior will follow a Dirichlet distribution:

$$\begin{aligned} p(\theta \mid x) &\sim \mathrm{Dir}(x_1+1, x_2+1, \ldots, x_d+1) \\ &= \prod_{j=1}^{d-1}(n+j)\left(\frac{n!}{\prod_{i=1}^{d} x_i!}\prod_{i=1}^{d}\theta_i^{x_i}\right)\cdot \end{aligned}$$

This implies that for the posterior potential we have

$$\begin{aligned} \varphi(\theta) &= \exp\left\{\sum_x \pi(x \mid \theta)\ln p(\theta \mid x)\right\} \\ &= e^{-\mathbb{H}[X|\theta]}\prod_{j=1}^{d-1}(n+j), \end{aligned}$$

where $\mathbb{H}\left[X \mid \theta\right]$ is the entropy of the multinomial distribution with parameter θ. The entropy of the multinomial distribution can be written in the following form:

$$\mathbb{H}\left[X \mid \theta\right] = -\ln(n!) - \sum_{i=1}^{d}\ln\phi(\theta_i),$$

where

$$\phi(\theta_i) = \exp\left\{n\theta_i\ln\theta_i - \sum_{j=0}^{n}\ln(j!)\binom{n}{j}\theta_i^j(1-\theta_i)^{n-j}\right\}\cdot \tag{32}$$

Therefore, for the posterior potential of the multinomial parameter we have

$$\varphi(\theta) = (n+d-1)!\prod_{i=1}^{d}\phi(\theta_i)\cdot \tag{33}$$

For the central probability measure of a multinomial model, we usually consider a query set that consists of random variables corresponding to fusions of the multinomial categories.

Let $X = (x_1, x_2, \ldots, x_d)$ be a random vector representing an element of the sample space of a multinomial model. A fusion random variable Y is obtained by summing up some entries of X. More precisely, if $\mathcal{A}_1, \ldots, \mathcal{A}_{d'}$ is a partition of the set $\{1, 2, \ldots, d\}$, each value of Y is a vector $(y_1, y_2, \ldots, y_{d'})$ where

$$y_j = \sum_{i\in\mathcal{A}_j} x_i, \qquad j = 1, \ldots, d'\ \cdot$$

The fusion variable Y also follows a multinomial distribution with the parameter $\omega = (w_1, w_2, \dots, w_{d'})$, where

$$w_j = \sum_{i \in \mathcal{A}_j} \theta_i, \qquad j = 1, \dots, d' \cdot$$

For example $(\theta_1 + \theta_5, \theta_2, \theta_4 + \theta_7 + \theta_8, \dots)$ fully parametrizes the fusion variable $(x_1 + x_5, x_2, x_4 + x_7 + x_8, \dots)$.

Suppose that in a multinomial model we aim to find the central probability measure for a query set $\mathcal{X} = \{X, Y_1, Y_2, \dots, Y_k\}$, where each Y_i is a fusion variable, and X represents the elements of the sample space. For using theorem 7, we need to obtain $\varphi(\omega_i)$, where ω_i is the parameter of Y_i, and also $\varphi(\omega)$, where $\omega = (\omega_1, \omega_2, \dots, \omega_k)$. Note that each ω_i is a multinomial parameter.

Each $\varphi(\omega_i)$ can be easily obtained using Equation 33. However, calculating $\varphi(\omega)$ requires integration:

$$\varphi(\omega) = \int_{\{\theta : \Omega(\theta) = \omega\}} \varphi(\theta) d\theta \cdot$$

This integration is performed over the set of all nonnegative solutions to the linear system of equations $\Omega(\theta) = \omega$. Here, Ω is a linear map with a matrix M_Ω consisting solely of 0 and 1 entries. The range of Ω typically has much lower dimensionality than θ, and the system of equations contains many free variables. This equation system has a solution when ω is inside the convex hull of the columns of M_Ω.

We convert this integration into an inference problem in a probabilistic graphical model, and then use a slightly modified version of the clique tree message passing [6] algorithm.

For representing equations of the linear system $\Omega(\theta) = \omega$ by a graphical model, we use indicator factors. Let $\vartheta \subseteq \{\theta_1, \theta_2, \dots, \theta_d\}$, and $0 \leq w \leq 1$. An indicator factor ξ is a function with the scope $\{\vartheta, w\}$ such that:

$$\xi(\vartheta, w) = \begin{cases} 1 & \sum_{\theta_i \in \vartheta} \theta_i = w, \\ 0 & \text{otherwise} \cdot \end{cases}$$

For example $\xi(\theta_2, \theta_5, \theta_6, w)$ is 1 when $\theta_2 + \theta_5 + \theta_6 = w$, and 0 otherwise.

Suppose the rank of M_Ω is r. We select r linearly independent equations from $\Omega(\theta) = \omega$, and define an indicator factor for every equation. Then, we construct an unnormalized measure ψ:

$$\psi(\theta, w_1, w_2, \dots, w_r) = \prod_{i=1}^{d} \phi(\theta_i) \prod_{k=1}^{r} \xi(\vartheta_k, w_k) \cdot \tag{34}$$

Clearly, marginalizing ψ over θ yields $\varphi(\omega)$. To achieve this, we construct a clique tree for ψ, denoted by $\mathcal{T}_\psi$, that has at least one clique C_w, containing all w_k for $k \in \{1, 2, \dots, r\}$. By performing message passing and selecting C_w as the root of the tree, we can compute the desired marginal $\psi(w_1, \dots, w_r)$ after summing out other variables in C_w.

In $\mathcal{T}_\psi$ each equation and each variable of the equation system $\Omega(\theta) = \omega$ is associated with a clique. Let C_i and C_j be two adjacent cliques in $\mathcal{T}_\psi$ with the sepset $\mathcal{S}_{i,j}$. Without loss of generality, we assume that the root C_w, is on the C_j side of the tree. Let $\mathcal{W}_{<(i,j)}$ be the set of all w_k for $k \in \{1, 2, \dots, r\}$, such that an indicator $\xi(\cdot, w_k)$ is associated with a clique on the C_i side of the tree. We can think of $\mathcal{W}_{<(i,j)}$ as the set of all equations that are associated with the C_i side of the tree. Let $\vartheta_{i,j}$ be the set of all θ_i for $i \in \{1, 2, \dots, d\}$ (i.e. the variables of the equation system) that are on the scope of both C_i and C_j. Due to the running intersection property, we have

$$\mathcal{W}_{<(i,j)} \cup \vartheta_{i,j} \subseteq \mathcal{S}_{i,j}.$$

However, during message passing, we can use a simplified sepset instead.

Definition 2.8. Let $\mathcal{T}$ be a clique tree. Let $\mathcal{R}_{i,j}$ denote the set of all possible assignments to the sepset $\mathcal{S}_{i,j}$. A *message reparameterization* for $\mathcal{T}$ is a map (i.e. function) $f_{i,j}$ defined from $\mathcal{R}_{i,j}$ to another set $\mathcal{R}^*_{i,j}$ for every i, j, which is message preserving. That means, for any $s_{i,j}, s'_{i,j} \in \mathcal{R}_{i,j}$, if $f_{i,j}(s_{i,j}) = f_{i,j}(s'_{i,j})$ then $\delta_{i\to j}(s_{i,j}) = \delta_{i\to j}(s'_{i,j})$, where $\delta_{i\to j}$ denotes the message from C_i to C_j.

The reparameterization of $\delta_{i\to j}$, is a function $\sigma_{i\to j}$ defined on $\mathcal{R}^*_{i,j}$ such that for every $s_{i,j} \in \mathcal{R}_{i,j}$, we have $\delta_{i\to j}(s_{i,j}) = \sigma_{i\to j}(f_{i,j}(s_{i,j}))$.

Proposition. We can replace all messages $\delta_{i\to j}$ with their reparameterization $\sigma_{i\to j}$ without affecting the message passing algorithm.

Proof. During message passing no information regarding specific assignments to $\mathcal{S}_{i,j}$ is passed between cliques. C_i simply produces $\delta_{i\to j}$ for every possible assignment to $\mathcal{S}_{i,j}$. Let Nb_i be the set of indexes of cliques that are neighbors of C_i. Having the mappings $f_{i,j}$ and $f_{k,i}$ for every $k \in Nb_i$, the reparameterized messages $\sigma_{k\to i}$ can be used to produce $\sigma_{i\to j}$. □

Suppose $w_k \in \mathcal{W}_{<(i,j)}$ and $\xi(\vartheta_k, w_k)$ is the indicator factor containing w_k. This indicator represents the equation

$$\sum_{\theta_i \in \vartheta_k} \theta_i = w_k.$$

We define a new variable $0 \leq w_k^{(i,j)} \leq 1$ for every $w_k \in \mathcal{W}_{<(i,j)}$ such that

$$w_k^{(i,j)} = w_k - \sum_{\theta_i \in \vartheta_k \cap \vartheta_{i,j}} \theta_i.$$

For each sepset $\mathcal{S}_{i,j}$, there will be a corresponding set of these new variables. We denote this set by $\mathcal{W}^*_{i,j}$. We have a map from assignments to $\mathcal{S}_{i,j}$ into assignments to $\mathcal{W}^*_{i,j}$, which is message preserving. If two assignments $s_{i,j}$, $s'_{i,j}$ to the sepset variables induce the same assignment to $\mathcal{W}^*_{i,j}$ variables, that implies the equation system corresponding to the C_i part of the tree has the same solution set for $s_{i,j}$ and $s'_{i,j}$. As a result, we will have $\delta_{i\to j}(s_{i,j}) = \delta_{i\to j}(s'_{i,j})$.

We can simplify message computation by introducing certain constraints on the structure of $\mathcal{T}_\psi$.

Definition 2.9 (Convolution Clique Tree)**.** Let $\mathcal{T}_\psi$ be a clique tree for the factorized measure of Equation 34. Let $pr(i)$, for each clique C_i, be the index of the parent of C_i, i.e. the neighbor on the path to the root clique C_w. We refer to $\mathcal{T}_\psi$ as a *convolution clique tree* if for every i we have

$$C_i - \mathcal{S}_{i,pr(i)} = \{\theta_i\}.$$

That is, in each clique exactly one variable is eliminated. We also require that each $\phi(\theta_i)$ factor be associated with the clique that eliminates θ_i.

Note that indicator factors may be associated with any clique as long as the tree remains family preserving.

Definition 2.10. In a convolution clique tree, a clique that is associated with at least one indicator factor is an *assignment node*. Any clique that is not an assignment node is a *convolution node*.

To have a more convenient notation, we can express messages as functions of vectors. Thus, the reparameterized message $\sigma_{i\to j}$ will be a function of a vector $w^*_{i,j}$, with an entry for each variable in $\mathcal{W}^*_{i,j}$. Note that each variable in $\mathcal{W}^*_{i,j}$ corresponds to an equation. Alternatively, $\mathcal{W}^*_{i,j}$ can be viewed as a set of equations. We define $\mathbf{1}_{i,j}$, for each i, j, to be an indicator vector indicating those equations of $\mathcal{W}^*_{i,j}$ that include θ_i (the variable that is elaminated in C_i).

If C_i is a convolution node, the outgoing message from C_i to C_j can be computed by convolution:

$$\sigma_{i\to j}(w^*_{i,j}) = \int \phi(t) \prod_{k\in Nb_i - \{j\}} \sigma_{k\to i}(w^*_{k,i} - t\mathbf{1}_{k,i})dt$$

On the other hand, when C_i is an assignment node, for every value of $w^*_{i,j}$ there is only one acceptable value for θ_i and the message can be computed without convolution. Moreover, certain values of $w^*_{i,j}$ will be invalid. For each indicator factor associated with C_i that does not include θ_i, the corresponding variable in $\mathcal{W}^*_{i,j}$ must always be 0 in the computed message. For those that include θ_i, the corresponding variables in $\mathcal{W}^*_{i,j}$ must be the same value and equal. If we denote this value by t, we have

$$\sigma_{i\to j}(w^*_{i,j}) = \phi(t) \prod_{k\in Nb_i-\{j\}} \sigma_{k\to i}(w^*_{k,i} - t\mathbf{1}_{k,i}).$$

Thus, in this equation the entries of $w^*_{i,j}$ that correspond to equations including θ_i are t, the entries corresponding to equations that do not include θ_i are 0, and other entries take any value.

A Expectation as a Linear Map

A sample space, denoted by $\mathcal{S}$, is the set of all possible outcomes of a random experiment. That is, the outcome of a random experiment can *always* be represented by *exactly one* element of $\mathcal{S}$.

Definition A.1 (Action Space)**.** Let $\mathcal{S}$ be a sample space. The action space of $\mathcal{S}$, denoted by $\mathcal{A}_\mathcal{S}$, is the set of all functions from $\mathcal{S}$ to $\mathbb{R}$, where $\mathbb{R}$ is the field of real numbers.

Proposition. The action space equals $\mathbb{R}^\mathcal{S}$, and is a vector space.

We can think of the members of $\mathcal{A}_\mathcal{S}$ as hypothetical actions or gambles. Each action associates an amount of loss with every possible outcome $s \in \mathcal{S}$. This loss is quantified by a real number. A negative loss indicates profit.

Definition A.2 (Sure Thing)**.** The *sure thing*, denoted by c, is an element of $\mathcal{A}_\mathcal{S}$, such that $c(s) = 1$ for every $s \in \mathcal{S}$.

Definition A.3. Let $\mathcal{E} \subseteq \mathcal{S}$. The restriction to $\mathcal{E}$ operator, denoted by $R_{\cap\mathcal{E}}$, is a linear operator on $\mathcal{A}_\mathcal{S}$, such that $R_{\cap\mathcal{E}}(\ell_\mathcal{S})$ for any $\ell_\mathcal{S} \in \mathcal{A}_\mathcal{S}$ is an element of $\mathcal{A}_\mathcal{S}$ and

$$R_{\cap\mathcal{E}}(\ell_\mathcal{S})(s) = \begin{cases} \ell_\mathcal{S}(s) & s \in \mathcal{E}, \\ 0 & s \in \mathcal{S} - \mathcal{E}. \end{cases}$$

We denote the range of $R_{\cap\mathcal{E}}$ by $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$.

Proposition. $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ is an invariant subspace of $\mathcal{A}_\mathcal{S}$ under $R_{\cap\mathcal{E}}$. If $\{\mathcal{E}_1, \mathcal{E}_2, \ldots\}$ is a partition of $\mathcal{S}$, then

$$\mathcal{A}_\mathcal{S} = \mathcal{A}_{\mathcal{S}\cap\mathcal{E}_1} \oplus \mathcal{A}_{\mathcal{S}\cap\mathcal{E}_2} \oplus \ldots,$$

where $\oplus$ indicates direct sum [2].

Definition A.4 (Linear Functional)**.** Let $\mathcal{V}$ be a vector space over a field $\mathbb{F}$. A linear functional on $\mathcal{V}$ is a linear map from $\mathcal{V}$ to $\mathbb{F}$.

Definition A.5 (Expectation)**.** An *expectation* for $\mathcal{S}$ is a linear functional on $\mathcal{A}_\mathcal{S}$, denoted by $\mathbb{E}_\mathcal{S}$, with the following properties:

- For the sure thing $c_\mathcal{S}$, we have $\mathbb{E}_\mathcal{S}\left[c_\mathcal{S}\right] = 1$.
- For any $\mathcal{E} \subseteq \mathcal{S}$, we have $\mathbb{E}_\mathcal{S}\left[R_{\cap\mathcal{E}}(c_\mathcal{S})\right] \geq 0$.

Proposition. $\mathbb{E}_\mathcal{S}$ is an element of the dual space of $\mathcal{A}_\mathcal{S}$.

Definition A.6 (Conditional Expectation)**.** Let $\mathbb{E}_\mathcal{S}$ be an expectation for $\mathcal{S}$ and let $\mathcal{E} \subseteq \mathcal{S}$. $\mathbb{E}_\mathcal{S}$ conditioned on $\mathcal{E}$, denoted by $\mathbb{E}_{\mathcal{S}|\mathcal{E}}$, is a linear map on $\mathcal{S}$ with the following properties:

- For the sure thing $c_\mathcal{S}$, we have $\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[c_\mathcal{S}\right] = 1$.
- For any $\ell_\mathcal{S} \in \mathcal{A}_\mathcal{S}$, and $\mathcal{E} \subseteq \mathcal{S}$, we have $\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[\ell_\mathcal{S}\right] = 0$ if and only if $\mathbb{E}_\mathcal{S}\left[R_{\cap\mathcal{E}}(\ell_\mathcal{S})\right] = 0$.

At this point, we have a complete framework for decision-making under uncertainty. Interestingly, we do not need to explicitly define probability. However, in order to present a more familiar framework, we provide a definition of conventional probability.

Definition A.7 (Probability)**.** We define the probability of $\mathcal{E} \subseteq \mathcal{S}$ as follows:

$$P(\mathcal{E}) = \mathbb{E}_\mathcal{S}\left[R_{\cap\mathcal{E}}(c_\mathcal{S})\right] \cdot$$

We now will establish the familiar relationship between conditional expectation and normal expectation.

Lemma 2. *Let $\mathcal{F}$ and $\mathcal{G}$ be two linear functionals defined on a vector space $\mathcal{V}$. $\mathcal{F}$ and $\mathcal{G}$ have the same null space if and only if $\mathcal{F} = \lambda\mathcal{G}$ for some $\lambda \neq 0$.*

Proof. If $\mathcal{F} = \lambda\mathcal{G}$, $\mathcal{F}\left[v\right] = 0$ if and only if $\mathcal{G}\left[v\right] = 0$ for every $v \in \mathcal{V}$. This proves the if part.

We prove the only if part by contradiction. Assume $\mathcal{F} \neq \lambda\mathcal{G}$. Then there exist $u, v \in \mathcal{V}$ such that

$$\mathcal{F}\left[u\right]\mathcal{G}\left[v\right] - \mathcal{F}\left[v\right]\mathcal{G}\left[u\right] \neq 0\cdot$$

Let $w = \mathcal{F}\left[u\right]v - \mathcal{F}\left[v\right]u$. Since $\mathcal{V}$ is a vector space $w \in \mathcal{V}$. $\mathcal{F}$ and $\mathcal{G}$ are linear. Hence, $\mathcal{F}\left[w\right] = 0$ and $\mathcal{G}\left[w\right] = \mathcal{F}\left[u\right]\mathcal{G}\left[v\right] - \mathcal{F}\left[v\right]\mathcal{G}\left[u\right] \neq 0$ and the null spaces of $\mathcal{F}$ and $\mathcal{G}$ are not equal. This contradiction completes the proof. $\square$

Theorem 9. *Let $\ell_{\mathcal{S}} \in \mathcal{A}_{\mathcal{S}}$ and $\mathcal{E} \subseteq \mathcal{S}$, where $\mathcal{S}$ is a sample space. If $R_{\cap\mathcal{E}}$ is the restriction operator to $\mathcal{E}$, then we have*

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[\ell_{\mathcal{S}}\right] = \frac{\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(\ell_{\mathcal{S}})\right]}{P(\mathcal{E})}\cdot$$

Proof. For an arbitrary $\ell \in \mathcal{A}_{\mathcal{S}}$, let $\ell' = \ell - R_{\cap\mathcal{E}}(\ell)$. Clearly, $R_{\cap\mathcal{E}}(\ell')$ is the zero function, and therefore $\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(\ell')\right] = 0$. Since $\mathcal{A}_{\mathcal{S}}$ is a vector space we have $\ell' \in \mathcal{A}_{\mathcal{S}}$, and the definition of conditional expectation implies $\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[\ell'\right] = 0$. Therefore

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[\ell\right] - \mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[R_{\cap\mathcal{E}}(\ell)\right] = 0\cdot \tag{35}$$

Let $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ represent the range of $R_{\cap\mathcal{E}}$. $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ is a subspace of $\mathcal{A}_{\mathcal{S}}$. By the definition of conditional expectation, functionals obtained by the restrictions of $\mathbb{E}_{\mathcal{S}}$ and $\mathbb{E}_{\mathcal{S}|\mathcal{E}}$ to $\mathcal{A}_{\mathcal{S}\cap\mathcal{E}}$ have the same null space. That implies by Lemma 2 for every $\ell \in \mathcal{A}_{\mathcal{S}}$ we have

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[R_{\cap\mathcal{E}}(\ell)\right] = \lambda\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(\ell)\right]\cdot$$

Thus, by Equation 35

$$\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[\ell\right] = \lambda\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(\ell)\right]\cdot$$

On the other hand, for the sure thing $c_{\mathcal{S}}$, we have

$$\lambda = \frac{\mathbb{E}_{\mathcal{S}|\mathcal{E}}\left[c_{\mathcal{S}}\right]}{\mathbb{E}_{\mathcal{S}}\left[R_{\cap\mathcal{E}}(c_{\mathcal{S}})\right]} = \frac{1}{P(\mathcal{E})}\cdot$$

This proves the theorem. □

Definition A.8 (Dual Map)**.** Let $\mathcal{A}$ and $\mathcal{B}$ be two sets. Suppose $M : \mathcal{A} \to \mathcal{B}$ is a map from $\mathcal{A}$ to $\mathcal{B}$. The dual map of M is the map $M^* : \mathbb{R}^{\mathcal{B}} \to \mathbb{R}^{\mathcal{A}}$, defined for each $f \in \mathbb{R}^{\mathcal{B}}$ by

$$M^*(f) = f \circ M\cdot$$

Proposition. The dual map is a linear map.

Theorem 10. *Let $\mathcal{S}$ be a sample set. Suppose $M : \mathcal{S} \to \mathcal{S}'$ is a map from $\mathcal{S}$ to another set $\mathcal{S}'$. If $\mathbb{E}_{\mathcal{S}}$ is an expectation for $\mathcal{S}$, then $\mathbb{E}_{\mathcal{S}'} = M^{**}(\mathbb{E}_{\mathcal{S}})$ is an expectation for $\mathcal{S}'$, where M^{**} is the dual map of M^*, and M^* is the dual map of M.*

Proof. M^{**} is a dual map and therefore is linear.

By definition for every $\ell_{\mathcal{S}'} \in \mathcal{A}_{\mathcal{S}'}$ we have

$$\mathbb{E}_{\mathcal{S}'}[\ell_{\mathcal{S}'}] = \mathbb{E}_{\mathcal{S}} \circ \ell_{\mathcal{S}'} \circ M \cdot$$

We can easily verify that

$$c_{\mathcal{S}'} \circ M = c_{\mathcal{S}} \cdot$$

Therefore, $\mathbb{E}_{\mathcal{S}'}[c_{\mathcal{S}'}] = 1$.

On the other hand for any $\mathcal{E}' \subseteq \mathcal{S}'$, we have

$$R_{\cap \mathcal{E}'}(c_{\mathcal{S}'}) \circ M = R_{\cap \mathcal{E}}(c_{\mathcal{S}}),$$

where

$$\mathcal{E} = \{s \in \mathcal{S} : M(s) \in \mathcal{E}'\} \cdot$$

This implies $\mathbb{E}_{\mathcal{S}'}[R_{\cap \mathcal{E}'}(c_{\mathcal{S}'})] \geq 0$. □

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